Multi-omics identifies key genetic and metabolic networks regulating spike organ development in wheat

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Short title: Exploring developmental metabolite networks in the 3-level structure of wheat spike.

ABSTRACT

Wheat (*Triticum aestivum* L.) spike development is tightly regulated by genetic and metabolic programs that drive organ growth and morphological changes. However, the dynamic interplay between metabolic shifts, gene expression patterns, and their regulatory roles during spike development, remains poorly characterized. To address this knowledge gap, we performed integrated metabolomic and transcriptomic profiling across 12 stages of wheat spike organ development. Our analysis detected 1,105 metabolites in 233 spike, spikelet, and floret samples, uncovering an uneven distribution of phytohormone-related metabolites. The exogenous phytohormone treatments validated the regulatory roles of phytohormones in spike morphogenesis. High-resolution spatiotemporal data from carpel organs enabled the reconstruction of a regulatory network, identifying key genes (including *12-oxo-phytodienoic acid reductase3* (*TaOPR3*), *Grain Length1* (*GL1*), and *Grain Length2* (*GL2*)) as critical determinants of grain size. Genomic analyses revealed geographical differentiation in gene haplotypes and their selective retention during breeding, with superior alleles associated with increased grain size. This comprehensive dataset provides a valuable resource for understanding the molecular basis of wheat grain yield and offers potential targets for crop improvement.

INTRODUCTION

Wheat (*Triticum aestivum* L.) is one of the three major global staple crops, requires genetic improvement and precision agronomy to enhance its yield for meeting worldwide food demands[1]. Optimizing spike architecture (particularly regulating spikelet fertility and grain arrangement) represents an effective strategy for yield improvement. The wheat spike is a compound spike inflorescence with alternately arranged spikelets along the rachis[2]. In hexaploidy wheat, a single spike typically bears 15-30 spikelets, each capable of developing 3-5 fertile florets[2,3]. Florets consist of structures including awns, lemma, palea, anthers, and carpels[2], with their developmental progression directly determining spike morphology and grain numbers [4,5].

Recent advances in multi-omics integration have accelerated the dissection of spike

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development mechanisms. Systematic transcriptome, chromatin accessibility, and histone modification analyses across eight critical developmental stages have revealed dynamic transcriptional networks governing spikelet formation[6]. Integrated metabolomic, transcriptomic, and genetic approaches have identified key genes controlling spike morphogenesis[7,8]. Metabolomics provides mechanistic insights into plant organ development, complementing limitations of genomic and transcriptomic studies[7,8]. The past decade has witnessed significant progress in wheat genomics, including high-quality genome assemblies[9-14], alongside widespread applications of QTL mapping and genome-wide association studies (GWAS) for investigating wheat development and abiotic/biotic stress resistance[15-18]. High-throughput metabolite profiling technologies have substantially enhanced metabolic phenotyping efficiency[19], th`ough current research predominantly focuses on mature grain composition[20,21], lacking systematic evaluation of metabolites across different developmental stages and spatial positions within wheat spikes.

Large-scale transcriptome analyses have systematically identified wheat spike regulatory modules, deepening understanding of floral organ development[22-25]. Bread wheat, an allohexaploid species (2n = 6x = 42, AABBDD), with genome derived from three distinct diploid progenitors, has revealed through transcriptomic studies how polyploidization influences gene expression, genomic imprinting, and heterosis—collectively regulating hexaploid wheat's spike morphology, grain size, and numbers [26-29]. However, no study has systematically profiled gene expression throughout wheat carpel development. While rice (*Oryza sativa* L.), tomato (*Solanum lycopersicum*), kiwifruit (*Actinidia chinensis cv* Hongyang) have established metabolic dynamics and transcriptional networks across life cycles[30-32], the transcriptome-metabolome integrative network during wheat carpel development remains unexplored. This study constructed spatiotemporal metabolic atlases for 233 wheat spike organs, uncovering metabolite distribution patterns across developmental windows. Organ-specific metabolic profiles provide a framework for deciphering complex biological processes involving multi-organ coordination. We established the most comprehensive carpel development transcriptome and metabolome dataset to date (covering 12 developmental stages), with multi-omics integration revealing: 1. Core genes and regulatory networks governing grain size. 2. Genetic signatures of geographical differentiation, breeding selection, and domestication. 3. Functionally significant non-synonymous SNPs/Indels in target genes. Notably, introgression of favorable alleles at these loci significantly increased grain size, confirming their yield-enhancing potential. This research establishes a resource foundation for investigating metabolic regulation of wheat growth, development, and nutrient accumulation.

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RESULTS

High-spatiotemporal-resolution dissection of spike components in wheat

To systematically capture metabolic dynamics during wheat spike development, we established the Wheat Spike Metabolomics Network Dataset—a resource that achieves high spatiotemporal-resolution monitoring of metabolic dynamics through metabolomic analysis (LC-MS/MS[19]) across 12 developmental stages and multiple organ types, with transcriptomic data (RNA-seq) integrated for carpel (Fig. 1A-D). Our sampling strategy encompassed: **1. Spatial Dissection**: 1) Spike-level: spikelets and rachis organs from five positions including top, upper middle (30%), middle, lower middle (70%), and bottom (Fig. 1A). 2) Spikelet-level: florets (F1–F4) and glumes within individual spikelets (Fig. 1B). 3) Floret-level: five floral organs-awn, lemma, palea, anther, and carpel (Fig. 1C). **2. Temporal Dissection**: samples were collected at 12 developmental stages: 1) Pre-anthesis: green anther (GrA), yellow anther (YA), tipping (TP), heading (HD). 2) anthesis (AN). 3) Post-anthesis: 5-35 days post-anthesis (DPA) in 5-day increments (Fig. 1D). Anthers were sampled only until AN, while other organs were tracked through 35 DPA.

Given the carpel's critical role in grain yield, we generated the most comprehensive metabolomic and transcriptomic dataset for this organ (Table S1), providing a foundational resource for studying spike development and metabolic regulation.

Developmental Dynamics of Spike Morphology

Comprehensive phenotyping generated 1,040 datasets that systematically documented spatiotemporal variations in spike, spikelet and floral organ (awn, lemma, glume, carpel) weights and spike length parameters across 12 developmental stages and 5 spatial positions (Table S2) During the developmental stages, a rapid increase in spike weight and grain weight was observed from AN to 35DPA (Fig. 1E, F). Spike length exhibited a sharp increase from GrA to TP, followed by a more gradual rise, peaking at the HD stage (Fig. 1G). Chaff weight increased at a higher

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rate during the TP-10DPA phase than during the 10DPA-35DPA phase (Fig. 1H). Analysis of spikelet weight, floret weight, awn weight, palea weight, glume weight, and carpel weight at different spike positions revealed that organ development occurred earlier in the middle region compared to the top and bottom regions (Fig. 1I, 1J, Table S2), supported by multiple lines of evidence: anthers in middle spikelets were dehisced at anthesis, while those in top and bottom spikelets remained green (Fig. S1A); there was a developmental gradient among florets (F1-F4) in middle spikelets, with F1 grains completed pollination and F4 stigmas still receptive (Fig. S1B). Additionally, dynamic changes in organs (including awn, glume, lemma, and carpel) within the middle florets indicated that F1 and F2 florets exhibited more advanced development than other florets (Fig. 1K, Supplementary Table 2). Meanwhile, the overall spike developmental stages were significantly correlated with weight ($r = 0.89$, $p<0.01$), with spikelets in the middle (with dehisced anthers) showing significantly higher weight than immature top spikelets ($p = 0.003286$), suggesting an association between developmental progression and biomass accumulation.

Spatiotemporal metabolite profiling

To gain a deep understanding of the dynamics of metabolites during spike developmental process, we used widely targeted tandem mass spectrometry (LC-MS/MS)[19] to determine the metabolites. A total of 1,105 metabolites were identified, of which 313 metabolites were annotated. These annotated metabolites included 33 flavonoids, 50 amino acid derivates, 30 lipids, 7 organic acids, 29 nucleic acid derivatives, 25 phenolamides, 38 phytohormone derivatives, 33 polyphenols, 9 sugars and 10 vitamins (Fig. 2A, Table S3). Analysis of this metabolites across different spike tissues revealed stage-specific metabolic patterns during spike development. However, metabolic profiles were similar among spikelets and florets from different positions, whereas significant differences existed between the rachis and spikelet tissues. Moreover, distinct metabolic patterns were also observed among different tissues within a single floret. For instance, metabolites such as flavonoids and phenolamides preferentially accumulated in the rachis, glumes, and lemma, palea, concomitant with increasing lignification; in contrast, levels of amino

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acid derivatives, lipids, nucleic acid derivatives, and vitamins were markedly higher in spikelet and floret tissues (Fig. 2A). Notably, despite the elevated abundance of these primary metabolites in reproductive tissues (spikelet/floret), their compositional ratios varied significantly depending on developmental stages, suggesting dynamic metabolic reprogramming during organ maturation. Flavonoids, phenolamides, and polyphenols showed strong correlations in lemma, glume, palea, and rachis organs. Similarly, phytohormone derivatives and polyphenols were highly correlated in spikelet, carpel, and floret organs (Fig. S2, Tables S4-S5). Principal component analysis revealed three distinct developmental phases: early phase (GrA-HD), transition phase (AN-5DPA), maturation phase (10DPA-35DPA) (Fig. 2B). Hierarchical clustering confirmed these developmental groupings (Fig. 2C).

Spatiotemporal Metabolic Signatures Reveal Tissue-Specific Roles in Wheat Spike Development

To dissect tissue-specific metabolic dynamics during wheat spike development, we profiled 313 annotated metabolites across 233 samples derived from spikelets, florets, carpels, and other organs spanning 12 developmental stages. Clustering analysis revealed nine distinct organ-enrichment patterns (Fig 3A). Notably, Cluster 5 was specifically enriched in anthers and contained 11 hormone-related metabolites, including cytokinin and auxin derivatives (Table S6). Clusters 3 and 6 were mainly accumulated in glume, lemma, and palea, whereas clusters 8 and 9 were enriched in spikelets, florets (F1–F4), and carpels (Fig. 3B; Table S3). These clusters reflected clear spatial signatures of metabolite distribution across different spike organs.

Hormone-associated metabolites peak in anthers and relate to developmental transitions

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Cluster 5 metabolites were significantly enriched in phytohormones and associated pathways, including ascorbate metabolism, zeatin biosynthesis, and plant hormone signal transduction. Among them, cytokinin dehydrogenase (CKX; EC 1.5.99.12) was identified as a key enzyme to regulate cytokinin degradation and anther development[33] (Fig. 3B, Table S7). Notably, several auxin conjugates (e.g., IAA-Asp, IAA-Trp, IAA-Glu) and oxidative products (OxIAA-Asp) peaked in abundance in anthers at the heading (HD) and anthesis (AN) stages (Figs. 3C, S3A).

Spatial distribution analysis revealed distinct auxin (IAA) accumulation patterns, showing pronounced enrichment in anthers during the HD stage, whereas carpels and spikelets exhibited elevated levels at 10 days post-anthesis (10DPA) (Fig. S3B). Notably, phytohormone, flavonoids, and lipid metabolites displayed anther-specific accumulation peaks during both HD and AN stages (Fig. S3C, S3D). These metabolites were functionally linked to key metabolic pathways, such as fatty acid elongation, purine metabolism, aromatic amino acid biosynthesis, and linoleic acid metabolism (Fig. S3E; Tables S8, S9).

Morphological observations showed that anthers expanded rapidly between the GrA and HD stages, followed by pollen grain maturation and release during the HD-AN stage (Figs. 3D, 3E). Correlation analysis indicated that cluster 5 metabolites (OxIAA, X5OHIAA, and IAA-Phe) were significantly positively correlated with anther size (Fig. 3G). This discovery strongly aligns with established auxin functions: in *Arabidopsis*, tomato and barley, IAA not only regulates inflorescence and floral development[34-36], but its transport disruption (e.g., PIP5K-mediated) also leads to pollen development defectst[37]. Notably in barley, the pollen-specific enzyme HvYUCCA4 synthesizes auxin to directly activate energy metabolism gene expression, thereby fueling maturation processes like starch accumulation[36]. Collectively, these findings reveal a conserved "metabolic reprogramming-energy

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supply" mechanism through which auxin regulates pollen development, providing direct evidence for IAA signaling in wheat anther development regulation.

Carpel-specific metabolic signatures point to sugar and nutrient transport pathways

Clusters 8 and 9 metabolites were specifically enriched in spikelet, F1-F4 floret, and carpel organs. These clusters comprised: 9 amino acid derivatives, 5 lipids, 2 nucleic acid derivatives, 3 sugars, 2 vitamins (Fig. 3B, Table S3).

Notably, cluster 9 metabolites exhibited dramatic accumulation at 20 days post-anthesis (20DPA) during carpel development (Fig. S4A). KEGG enrichment analysis highlighted starch and sucrose metabolism histidine metabolism, and ABC transporters (Fig. S4B, Table S7). β-alanine metabolism, Within the starch and sucrose metabolism pathway (ath00500), which contains key regulatory enzymes including β-fructofuranosidase (EC 3.2.1.26), α-glucosidase (EC 3.2.1.20), and sucrose-phosphate phosphatase (EC 3.1.3.24) (Table S7). Through BLASTp analysis of amino acid sequences derived from wheat gene indices (ats) in KEGG, 14 wheat homologs encoding these enzymes were identified, among which *grain incomplete filling 1* (*TaGIF1*) was screened as a candidate gene with the highest alignment rate (Table S10). *TaGIF1* is a known regulator of grain development. Comparative analysis revealed functional conservation of *GIF1* homologs in *Arabidopsis* and rice, where they mediate grain morphogenesis, organ size regulation, and iron absorption[38-42].

TaGIF-D1 (*TraesCS2D02G293200*) was highly expressed in developing spike and grain organs (Fig. S4C, Table S11). Intriguingly, *TaGIF-D1* expression exhibited a significant negative correlation with sucrose and cellotriose levels in carpel organ, suggesting that it may participate in modulating sucrose partitioning during grain filling (Fig. S4D). These findings parallel the established mechanism in rice, where *GIF1* regulates sucrose transport to control grain filling efficiency.

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The spatial distribution differences of hormones significantly affect spike morphology

Through analyzing 313 metabolites across 12 developmental stages and 5 spikelet positions, we identified six temporally distinct metabolic modules (Sp_cluster1-6), each displaying characteristic stage-specific expression peaks (Fig. 4A-B). Importantly, Sp_cluster1, 3, 5, and 6 were significant enrichment in plant hormone signal transduction pathways (Fig. 4C, Table S12), suggesting the involvement of hormonal networks in the temporal regulation of spikelet development.

Building on this finding, we systematically analyzed spatial differences in hormone-related metabolites among five spikelet positions using Fisher's LSD multiple comparisons. The results revealed significant spatial distribution differences (present in ⩾2 developmental stages) for metabolites including 2-oxindole, abscisic acid (ABA), isopentenyl adenosine (IP), salicylic acid (SA), IAA-ALA, IAM, and OxIAAsp (Fig. 4D, Tables S13-S14), demonstrating the natural heterogeneity of endogenous hormone levels within spike organs.

To investigate the effects of hormonal signals on spikelet development, we conducted systematic exogenous hormone treatments from the GrA to 35 DPA stages (Fig. 4E-H). While whole-plant spraying cannot precisely replicate endogenous spatial distribution patterns, it effectively validates the responsiveness of spike tissues to hormonal signals. The experimental results demonstrated significant position-specific responses to hormone treatments (Fig. 4E-H): Cytokinin (CK) application specifically promoted the development of basal spikelets, as evidenced by a dramatic 96.97% reduction in abortion rate ($p<0.001$) and an increase of 3.33 fertile spikelets number (Table S15). Furthermore, the number of florets per spikelet in bottom spike position increased by 3.14 numbers, accompanied by a significant 21.97% increase in grain number (2.17 more grains, $p < 0.01$) (Table S16). In contrast, ABA treatment exhibited inhibitory effects across all spikelet positions,

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resulting in an overall 35.31% decrease in dry weight ($p<0.05$) and a reduction of 11.7 grains per spike ($p<0.01$) (Table S16). Notably, the middle and down position spikelets showed particularly pronounced sensitivity to ABA, with significantly decreased grain number per individual spikelet (Table S16). SA, IAA, and JA treatments selectively decreased grain number and weight in top position spikelets (Table S16), and GA application caused top spikelet thinning and degeneration compared to wild type (Fig. 4E-H). These findings reveal that spikelets at different positions may possess intrinsic and position-specific hormonal response mechanisms. Future studies should employ more precise spatial manipulation techniques to directly validate the causal relationship between endogenous hormone distribution patterns and spike development.

Integrated transcriptomic and metabolomic analyses reveal key regulatory networks governing grain development and size in wheat

To elucidate the molecular regulators of carpel development, we constructed a co-expression network using transcriptome and metabolome data from 12 developmental stages. This analysis identified 42,435 genes co-regulated with at least one metabolite (Table S17). Genes and metabolites were grouped into six co-response clusters (I–VI) (Fig. 5A, 5B; Tables S3, S18).

Cluster I was enriched for functions related to nutrient storage (*e.g.*, α-amylase inhibitor activity), starch biosynthesis, seed development, and carbohydrate metabolism (Tables S19–S21), suggesting its role in supplying energy and substrates for carpel growth. Leveraging the Plant Metabolic Network (PlantCyc) and prior

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research[43], we identified 89 starch synthesis-associated genes in Cluster I, including *starch branching enzymes* (*TaSBEI*), *sucrose synthase* (*Susy 5*, *TaSuSy5*), *granule-bound starch synthase I* (*TaGBSSI*), *starch synthase* (*TaSSlc*), *ADP-glucose pyrophosphorylase large subunit 1* (*TaAGPL1)*, *isoamylase 1* (*TaISA1*), and *pullulanases* (*TaPUL*). Notably, *TaAGPL-D1* (*TraesCS1D02G427400*) showed the highest expression (Table S22). This gene encodes ADP-glucose pyrophosphorylase, a rate-limiting enzyme in starch biosynthesis; its overexpression enhances starch accumulation in wheat[44]. We also detected *trehalose-6-phosphate phosphatase* (*TaTPP-7A*, *TraesCS7A02G180800*) in Cluster I, which increases grain size when overexpressed[45].

In cluster III, transcriptome data revealed significant enrichment of KEGG terms related to grain size and hormonal regulation (Table S20). Within these pathways, we identified 49 auxin-related genes, including *PIN-FORMED 1*(*TaPIN1*), *TaARF12*, and *TaIAA21* (Table S23), which are known regulators of grain size and spike development in wheat[46-48]. Notably, these auxin-related genes exhibited strong positive correlations with phytohormone derivatives such as IP, indole, 2-oxindole, iPRMP, and BA9G (Fig. 5C, Table S23), indicating a functional interplay between auxin signaling and hormone metabolism in grain development. These results demonstrate the biological relevance of our clustering approach and highlight key molecular mechanisms underlying grain development.

Metabolic pathway analysis revealed that multiple pathways co-occurred within individual clusters (Tables S19-S21), demonstrating the complex coordination of metabolic processes during distinct developmental stages. Notably, we observed cluster-specific pathway enrichment patterns: phenolamide biosynthesis was prominent in clusters I and III, suggesting its functional importance during early-to-intermediate developmental stages. Flavonoid-associated metabolites were

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predominantly enriched in clusters II, IV, and VI, implicating their roles in later developmental phases (Fig. 5B).

These findings delineate stage-specific metabolic reprogramming during wheat development. By mapping these dynamic pathway activities to developmental transitions, our work provides a framework for investigating the metabolic regulation of wheat growth, offering targets for future mechanistic studies.

To identify key transcription factors (TFs) associated with metabolic regulation during grain development, we analyzed 2,066 TFs from the carpel development transcriptome using PlantTFDB (http://planttfdb.gao-lab.org/index.php?sp=Tae) (Table S24). Cluster I was enriched for heat stress transcription factors (HSFs) and NAC family TFs, while cluster III contained diverse regulators including ABI3/VP1, AP2/EREBP, ARFs, bHLH, TCP, YABBY, GRAS, MADS II, MYB-related, SBP, and WRKY TFs (Table S24), suggesting specialized roles in metabolic pathway regulation.

Correlation analysis revealed strong positive ($r > 0.85$) or negative ($r < -0.85$) relationships between TFs and metabolites. Key TFs associated with carpel development included: *SEP3*-like MADS-box (*TaSEP3*[49]), *TaARF12*, *TaARF25*[50], *AGAMOUS LIKE 6* (*TaAGL6*[51]), *BRASSINAZOLE-RESISTANT1* (*TaBZR1*[52]), *Reduced height-B1b* (*Rht-B1b*[53]).

Notably, cluster I metabolites (raffinose, cellotriose, alanine) showed negative correlations with cluster III TFs (ARF, AP2/EREBP, MADS II, BES1) (Fig. 5D, Table S3), likely reflecting a growth-stress tradeoff where stress-responsive metabolite accumulation coincides with growth pathway suppression (Tables 19-20, Fig. 5A).

WGCNA of carpel development (GrA to 35DPA) identified 10 co-expression

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modules (9,636 genes, CV ≤ 25%) (Fig. S5A, Table S25). The greenyellow module, positively correlated with seed dimensions (Fig. S5B), contained 21 low molecular glutenin genes (19 in cluster I). Our network analysis highlighted strong positive associations between *Squamosa Promoter Binding Protein-like* (*TaSPL16, TraesCS7A02G260500*, *TraesCS7B02G158500*, T*raesCS7D02G261500*) alleles and hormones (Indole, tZOG, IP, iPRMP) (Tables S17, S25), consistent with *TaSPL16*'s known role in enhancing grain size[54]. These results establish a comprehensive regulatory map linking metabolic pathways, TFs, and phenotypic outcomes during wheat grain development.

To elucidate the regulatory network governing grain/carpel development in wheat, we analyzed the transcriptome and metabolome datasets, focusing on co-response cluster VI, which exhibits high gene expression and metabolite accumulation at the AN stage. Integrated analysis revealed the α-linolenic acid metabolism pathway—critical for jasmonic acid (JA) biosynthesis (Table S19, S20). Gene Ontology (GO) enrichment of cluster VI transcriptome data highlighted JA-associated processes, including JA-mediated signaling, response to JA, and JA hydrolase activity. Key JA biosynthesis genes like *lipoxygenases* (*LOX)*, *allene oxide synthase* (*AOS)*, *allene oxide cyclas1* (*AOC1)*, and *12-oxo-phytodienoic acid reductase3* (*OPR3*) (Fig. 6A) were identified, with prior studies linking *OPR3* mutations to male sterility in *Arabidopsis*, wheat, and rice[55-61] .

In our previous research, *TaOPR3* (*TraesCS7A02G412400*, *TraesCS7B02G311600*, *TraesCS7D02G405500*) was shown to modulate the jasmonic acid (JA) pathway, enhancing heat tolerance in *Arabidopsis* and wheat[61]. However, its role in grain development remained uncharacterized. Notably, *TaOPR-A3* displayed higher expression in carpels compared to *TaOPR-B3* and *TaOPR-D3* at AN stage (Table S26). To investigate *TaOPR3*'s contribution to grain morphology, we generated overexpression (*OE-1*, *OE-2*, *OE-4*) and RNA interference (RNAi-1, RNAi-2, RNAi-3) lines of *TaOPR-A3*[61]. The RNAi lines exhibited reduced grain length and width relative to wild-type (KN199), while overexpression lines showed no

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significant phenotypic changes (Fig. 6B–E).

Sequencing of *TaOPR-A3* in two Chinese wheat accessions (Jingshuang16 and Bainong64) revealed a nonsynonmous SNP (+230 protein sequence, Arg/Gly). To assess its functional impact, we developed near-isogenic lines (NILs) by introgressing the *TaOPR-A3-Gly* allele from Bainong64 into Jingshuang16 (genetic background equivalent to Chinese Spring). Compared to the parental Jingshuang16 (*TaOPR-A3*-Arg, grain length:7.24mm; grain width:3.44mm), NILs harboring the *TaOPR-A3*-Gly allele showed increased grain dimensions (length:7.63mm for NIL1$^{TaOPR\text{-}A3\text{-}Gly}$, 3.56mm for NIL1$^{TaOPR\text{-}A3\text{-}Gly}$; width:7.87mm for NIL1$^{TaOPR\text{-}A3\text{-}Gly}$, 3.79mm for NIL2$^{TaOPR\text{-}A3\text{-}Gly}$) (Fig. S6A,B,Table S27).

These results demonstrate that *TaOPR3,* particularly the *TaOPR-A3* isoform, regulates grain size in wheat. The Arg→Gly substitution significantly influences grain morphology, as validated by NIL comparisons, suggesting *TaOPR3* as a promising target for wheat grain improvement.

Gene Ontology (GO) analysis of cluster VI identified two cytokinin-related pathways (Tables S18, S21). Previous studies in *Arabidopsis* and rice have demonstrated that cytokinin signaling is modulated by KISS ME DEADLY (KMD) F-box proteins, which regulate transcriptional responses through targeted degradation of transcription factors[62,63]. Within this cluster, we detected 43 F-box-related genes, including four homologs of *Oryza* sativa *OsKMD4* (*TraesCS4B02G087600*, *TraesCS4B02G087800*, *TraesCS4D02G084900*, *TraesCS4D02G085000*). Notably, *TraesCS4D02G085000* exhibited markedly higher expression in spikes and grains compared to its homologs (Fig. S7). Functional characterization revealed that overexpression of *TraesCS4D02G085000* significantly reduced grain length without affecting width (Fig. 6F-I). Given this phenotype, we designated this gene as *Grain Length1* (*GL1*).

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KEGG pathway analysis of Cluster III identified several key regulators of seed development and size, including *TaARF21*, *TaPIN1*, *TaIAA21* (Table S18, S20). This cluster was particularly enriched for ATP-dependent RNA helicase genes (Table S18), among which *TraesCS6A02G416700* showed high homology to known seed development regulators: *AtRH02* (*At3g19760*), and *AtRH34* (*At1g51380*) in *Arabidopsis*, and *OsRH2* (*LOC_Os03g36930.1*), *OsRH34* (*Os01g45190.1*) in rice[64-66]. We developed two mutants (mut-1 and mut-2) based on Kronos (wild type, a tetraploid wheat accession). Phenotypic analysis showed that grain lengths of *GL2-6A* mutant lines were increased by 6.91% to 7.96% in the greenhouses, respectively, compared with the wild-type Kronos. But no significant difference was observed in grain width between mutants and wildtype (Figure 6J-M). Based on this phenotype, we designated *TraesCS6A02G416700* as *Grain Length2* (*GL2*).

To further explore the potential contribution of *GL2* to grain size, we sequenced the *GL2* in two Chinese wheat accessions (Jingshuang16 and Bainong64). We detected a 15bp insertion (116 bp upstream of the ATG start codon) in Bainong64 relative to Jingshuang16. To determine the effect of this indel on grain size, we constructed NILs that carry the *GL2*-indel from Bainong64 through repeated backcrossing into Jingshuang16 which same as Chinese spring. Compared to the parental Jingshuang16 line (grain length:7.24mm), the improved Jingshuang16 lines carrying the *GL2*-indel had longer grains: grain length (8.11mm for NIL1$^{GL2\text{-indel}}$, 8.15mm for NIL2$^{GL2\text{-indel}}$, 7.97mm for NIL3$^{GL2\text{-indel}}$) (Fig. S6C, D, Table S27).

Comprehensive Analysis of *TaOPR3* Haplotype Variation in Wheat Populations

1. Population Structure and Study Design

We systematically analyzed *TaOPR3* variation across four distinct wheat populations: 1) **Global diversity panel**, 306 worldwide accessions (landraces and traditional cultivars). 2) **Regional comparison**, 355 Chinese and USA wheat accessions. 3) **Breeding history**, 829 Chinese wheat accessions released between 1900 and 2020. 4) **Domestication analysis**, 172 hexaploid accessions and 242 additional accessions (121 *Aegilops tauschii*, 121 tetraploid wheat).

2. *TaOPR-A3* Haplotype Analysis

2.1 Global Distribution Patterns In the worldwide panel, we indicated four major *TaOPR-A3* haplotypes: Hap-1$^{TaOPR\text{-}A3}$ (28.10% frequency), Hap-2$^{TaOPR\text{-}A3}$ (53.92%,

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matches Chinese Spring reference), Hap-3$^{TaOPR\text{-}A3}$ (5.88%) and Hap-4$^{TaOPR\text{-}A3}$ (10.13%) (Fig. 7A, Table S28). These haplotypes showed distinct geographical distributions (Fig. 7B, Table S28) and significant differences in thousand grain weight (TGW): Hap-3$^{TaOPR\text{-}A3}$:36.49g (lowest), Hap-1: 41.41g - Hap-2: 40.64g, Hap-4: 41.79g (highest) (Fig. 7C, Table S29)

2.2 Breeding Selection in China Analysis of 829 Chinese breeding accessions revealed: 55 SNPs defining 4 haplotypes. Progressive increase in non-reference haplotypes during breeding, Hap-c4$^{TaOPR\text{-}A3}$ (reference) frequency decreased over time. This suggests strong selection pressure on *TaOPR-A3* in Chinese breeding programs. (Fig. 7D. Table S30).

2.3 China-USA Comparison In 355 accessions, we observed: Hap-cu1$^{TaOPR\text{-}A3}$ predominated across categories, Hap-cu2$^{TaOPR\text{-}A3}$ more frequent in landraces (13.89%) vs cultivars (CHN-CV: 1.00% and USA-CV: 0.00%). Hap-cu3$^{TaOPR\text{-}A3}$ showed regional divergence (China: 17.00%, USA: 6.94%) (Fig. 7E, Table S31).

3. Domestication History Phylogenetic network analysis revealed: Hexaploid wheat divided into 4 haplotypes (Hap-IV-IX), distinct haplotypes in wild progenitors: Hap-I: einkorn wheat, Hap-II/III: urartu Hap-VII/VIII: emmer wheat. Haplotype similarity between hexaploid and tetraploid wheat suggests tetraploid origin (Fig. 7F).

For *TaOPR3-B3* (*TraesCS7B02G311600*), we identified three haplotypes: Hap-1$^{TaOPR\text{-}B3}$ (61.76%, 189 of 306 accessions), Hap-2$^{TaOPR\text{-}B3}$ (15.03%, 46 of 306 accessions) and Hap-3$^{TaOPR\text{-}B3}$ (23.20%, 71 of 306 accessions) in the 306 worldwide wheat accessions. Hap-1$^{TaOPR\text{-}B3}$ had the same sequence as the reference genome (Chinese Spring) (Fig. S8A, Table S32). The three major haplotypes of *TaOPR-B3* displayed worldwide geographical differentiation. Hap-1$^{TaOPR\text{-}B3}$ accounted for relatively high percentages in most regions (57.14-76.74%) except Oceania and South America, Hap-2$^{TaOPR\text{-}B3}$ was more widely distributed in Oceania (41.17%) and South America (30.95%) and not found in Africa, Hap-3$^{TaOPR\text{-}B3}$ was more widely

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distributed in USA and Europe, and less prevalent in Middle East (4.65%) (Fig. S8B, Table S32). In addition, we explored the history of *TaOPR-B3* haplotypes in 829 Chinese accessions. Hap-c1$^{TaOPR-B3}$ had the same sequence as the reference genome (Chinese Spring) and decreased in frequency from 80.17% to 45.91%, while other two haplotypes all increased (Hap-c2$^{TaOPR-B3}$:16.37 to 44.90%; Hap-c3$^{TaOPR-B3}$:3.45% to 18.18%) (Fig. S8C, Table S33). For 355 Chinese and USA accessions, five major haplotypes were identified, in which the percentage of Hap-cu1$^{TaOPR-B3}$ was higher in Chinese cultivars (CHN-CV) (54.37%) than in landrace (7.82%) and USA-CV (18.06%) categories. Hap-cu2$^{TaOPR-B3}$ was the major haplotype in landrace and USA-CV, Hap-cu3$^{TaOPR-B3}$ and Hap-cu4$^{TaOPR-B3}$ were only detected in landrace and CHN-CV, Hap-cu5$^{TaOPR-B3}$ was only identified in landraces (Fig. S8D, Table S34).
For *TaOPR3-D3* (*TraesCS7D02G405500*), we identified five, four and four main haplotypes in 306 worldwide accessions, 829 Chinese accessions and 355 Chinese and USA accessions, respectively. These haplotypes displayed differences in geography and breeding selection (Fig. S8E-H).

DISCUSSION

Enhancing grain yield represent a primary objective in modern wheat breeding programs, with spike architecture serving as fundamental determinant of yield potential. The hormonal regulation of inflorescence development in cereal crops combines conserved and species-specific patterns. Maize (*Zea mays*), barley (*Hordeum vulgare*), and rice (*Oryza sativa*) all utilize IAA and CK gradients to regulate inflorescence architecture, demonstrated by: 1) the spatiotemporal expression of IAA/CK/GA/JA-related genes in maize ears[67]. 2) the conserved auxin and cytokinin concentration gradient in barley spikes, higher at the base than the apex, which is critical for architecture[68]. 3). OsPID-mediated auxin transport controlling rice floral organ development[69]. Nevertheless, interspecies divergences exist: maize employs dynamic JA/ABA/SA/ethylene signaling for sexual differentiation[70]. while barley exhibits apical ABA accumulation triggering programmed tip degeneration (PTD), with auxin/cytokinin promote basal growth[71] Metabolites, particularly hormones, play crucial regulatory roles in spike morphology, influencing not only yield but also environmental adaptation and stresses resistance. Our findings demonstrate that spatiotemporal hormone distribution patterns regulate spikelet organogenesis through metabolic gradient establishment. The observed "position-specific hormone perturbation → morphological response" relationship provides a theoretical framework for optimizing spike architecture in crop improvement. To systematically understand these relationships, we generated a comprehensive spatiotemporal metabolome atlas

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of wheat spike development, mapping metabolite distributions across distinct spike components and twelve developmental stages using high-resolution metabolomics. Our study uniquely integrates morphological data with single-metabolite distribution patterns, revealing both spatial and temporal metabolic dynamics. Key findings demonstrate that: 1) Identical metabolites exhibit organ-specific concentration gradients within the same developmental stage 2) Individual metabolites show temporal fluctuation patterns within specific organs 3) These spatiotemporal variations create complex metabolic networks coordinating spike development. Notably, multi localizing metabolites appear to function through: Spatial confinement of activity within specific compartments, Stage-dependent concentration modulation enabling pleiotropic regulation

This systems-level perspective facilitates identification of hub metabolites governing spike-wide developmental networks. Supporting previous reports of gibberellin (GA) transport through vascular systems[72,73] , we observed substantial GA accumulation in rachis post-anthesis, suggesting dual roles in transport and storage.

The carpel is the key organ for grain yield in wheat. Previous studies reported the gene expression profiles for several stages after anthesis[22,43,74,75], but few works display the overall information of gene expression from initiation of grain/carpel to the maturity.

We constructed the most comprehensive transcriptome and metabolome dataset for carpel development. We demonstrated a network by integrating of the temporal metabolome and transcriptome data. Based on the network, we identified the critical genes regulating grain size in wheat, confirming its effectiveness.

Furthermore, we established the most complete transcriptome-metabolome dataset for carpel development, enabling network-based identification of key grain size regulators. Building upon recent genomic advances[9-14, 76-79]., we characterized haplotype distributions of three target genes (*TaOPR3*, *GL1*, *GL2*) across global geographical populations, century-long Chinese breeding history and diploid-to-hexaploid wheat domestication(Fig7,S8-9 Table S28-38).

Collectively, our findings support that spatiotemporal dynamics of hormone-related metabolites and the position-specific developmental responses elicited by exogenous hormone application underscore the critical importance of localized hormone levels and their spatial imbalances in regulating spikelet development and grain setting. In summary, we have established a high-resolution metabolomic atlas of wheat spike development that reveals insights into the spatiotemporal regulation of spike morphology. Our integrated analysis demonstrates the utility of these datasets for investigating multiple facets of grain yield determination. This comprehensive

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resource enables: 1) Systematic hypothesis generation regarding metabolic control of spike architecture. 2) Validation of spatial-temporal metabolic networks governing spikelet and floret development. 3) Identification of key genetic regulators of grain size through combined transcriptome-metabolome analysis

Furthermore, our findings establish a framework for comparative studies across cereal species to elucidate: Evolutionary conservation of developmental regulation networks. Species-specific metabolic adaptations in spike development.

MATERIALS AND METHODS

Plant materials and Sampling

The wheat cultivar Longchun35 (*Triticum aestivum* L.) were cultivated under field conditions at the Institute of Botany, Chinese Academy of Sciences (Beijing, China). Samples were collected from the green anther stage (GrA), yellow anther stage (YA), tipping stage (TP), heading stage(HD), anthesis (AN), and 5–35 days post-anthesis (DPA) in 5-day increments (DPA), with developmental staging following established protocols[80].

Spatial Sampling Protocol:

1. Spike Partitioning: Each spike was divided longitudinally into five equal segments: 1) the top of spike (top), 2) 30% of spike length (upper middle), 3) the center of spike (middle), 4) 70% of spike length (lower middle), 5) the bottom of spike (bottom). The samples of spikelet and rachis were collected at the five positions.

2. Organ Collection:

1) Spikelets and rachis were sampled from all five positions. 2) Central spikelet components were dissected into: florets(F1-F4), glumes. 3） F1 florets were further separated into: anthers, lemma, palea, awns and carpel (used for transcriptome and metabolomics analysis). Samples from more than ten individual plants were pooled

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as one biological replicate and immediately frozen in liquid nitrogen. Two biological replicates were used for metabolome and three biological replicates were used for transcriptome analysis, respectively.

The experiment was conducted in a test field with uniform soil conditions. To avoid marginal effects, all samples were collected from the center of the field (at least 0.5 meters from the edge). Ten individual plants were randomly selected as samples at each developmental stage, and the spacing between the sampled plants was at least 30 cm to ensure sample independence. All samples were collected at 9:00-11:00 a.m. (in the morning) during the day.

Phenotype

Spike morphology traits including spike length, spike weight, grain weight, chaff weight, rachis weight, spikelet weight at different stages, were measured following our previous established protocol[81]. Each trait was measured using samples collected from four distinct individual plants, which served as biological replicates. These individual plants were randomly selected from the experimental population to ensure that the replicates were independent and representative of the overall population variability.

During the 12 stages of spike development, we also measured the glume weight and the weights of different floret organs (lemma weigh, awn weight, and carpel weight) at five spike position. For these measurements, the biological replicates were again four separate individual plants.

Images of the samples were captured using a Stereo microscope (SZ810, CNOPTEC, Chongqing, China). Anther length, anther width, grain length and grain width were measured using ImageView software based on the microscope images.

For measuring the anther length and width, ten individual plants were utilized as biological replicates. Given that anther characteristics can exhibit significant variation among plants, a larger number of replicates was employed to increase the precision of the measurements and enhance the reliability of the statistical inferences. In contrast, for measuring grain length and grain width, three individual plants were used as biological replicates. Ten biological replicates were used for grain length and grain width.

Preparation of metabolic samples

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Samples were first freeze-dried and ground into powder using a mixer mill (MM400, Retsch) for 50 s at 30 Hz. For each sample, 50 mg fine-dry powder was weighed and extracted by 20 min sonication after 0.5 ml of 70% methanol w/v containing internal standards (0.1 mg/l acyclovir) was added. The supernatants were collected and passed through filter (SCAA-104, 0.22 mm pore size; ANPEL Shanghai, China, www.anpel.com.cn/) after centrifugation for 10 min at 10,000 g.

Metabolome profiling analysis

The metabolome analysis was generally followed the previously developed LC-MS/MS-based widely targeted metabolic profiling method[82]. To be specific, high resolution MS data were obtained in an untargeted metabolite analysis, the mixed samples were analyzed using a time–of–flight mass spectrometry platform (HPLC, Shim-pack UFLC SHIMADZU CBM20A system, Triple TOF 5600; Applied Biosystems). The identification of metabolites was performed based on comparisons of the obtained MS/MS spectra, exact mass numbers, and retention times with those of authentic standards and putatively annotated compounds that described in previous studies[20,21,83,84]. The relative levels of metabolites in samples were analyzed by an LC-ESI-MS/MS system (Shimadzu CBM20A UFLC system, 5500 QTRAP; Applied Biosystems, Framingham, MA, USA). An internal standard and a quality control sample were used to monitor detection performance. The sMRM algorithm was used with an MRM detection window of 90 s and a target scan time of 1.0 s using Analyst 1.6 software. The components were separated using a C18 column (Accucore C18, 150 × 2.1 mm; Thermo Scientific, Waltham, MA, USA). The mobile phases were 0.1% formic acid in water (buffer A) and acetonitrile (buffer B). The flow rate was set to 0.35 mL per min, and the following operational gradient was used: 0 min, 5% B; 20 min, 95% B; 22 min, 95% B; 22.1 min, 5% B; 25 min, 5% B. The injection volume was five μl and the column temperature was set to 40 °C. Metabolite identification was performed according to previously published methods[84-86].

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RNA isolation and sequencing

Total RNA was isolated from the carpel of Longchun35 (a hexaploid wheat accession). Total RNA was extracted using TRIzol reagent (Invitrogen), treated with DNase I, and purified using RNeasy columns (QIAGEN). One μg total RNA per sample was used as input material for the preparation of sequencing libraries using a NEBNext UltraTM RNA Library Prep Kit for Illumina (NEB, USA) following the manufacturer's recommendations. Index codes were added to attribute sequences to each sample following sequencing and demultiplexing. The clustering of index-coded samples was performed on a cBot Cluster Generation System using a TruSeq PE Cluster Kit v3-cBot-HS (Illumina) according to the manufacturer's instructions. After cluster generation, the libraries preparations were sequenced on an Illumina Novaseq 6000 platform and 150-bp paired-end reads were generated (Novogene, Beijing, China).

Read mapping and transcript profiling

The Chinese spring reference genome (IWGSC Annotation v1.1) was downloaded from http://ftp.ensemblgenomes.org/pub/plants/release-46/fasta/triticum_aestivum/dna/Triticum_aestivum.IWGSC.dna.toplevel.fa.gz. Raw RNA-seq reads were processed using Trimmonmatic[87] to remove adapter and low-quality sequence, and salmon[88] was used to map the reads to wheat reference genome. Following alignments, raw counts for each wheat gene were normalized to transcripts per million (TPM) by using R package tximport (v1.18.0)[89]. We calculated the Pearson correlation coefficient between three biological replications with the $\log_2$(TPM+1) and replaced the samples with low correlation with the mean of both samples. The transformed and normalized gene expression values with

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$\log_2$(TPM +1) were used for hierarchical clustering, and the z-scores of the genes were used for the analysis of PCA.

Metabolic Correlation and Enrichment Pattern Statistics

Metabolic data were $\log_2$(metabolic expression values +1) transformed to improve normality and homogeneity, and then got the normalized metabolic expression values with Z-score. Z-score were used for hierarchical clustering and PCA by R software. Pearson correlation coefficient was calculated on Z-score to estimate inter-organ and intra-organ metabolite pair correlations.

Metabolite Clustering Analysis

To systematically analyze the clustering patterns of metabolites across different samples and organs, we performed metabolite clustering analysis. This analysis was divided into two categories: one based on metabolites across all samples and the other based on metabolites within the same organ types, including spike, rachis, floret, and ovary. In each clustering analysis, we first applied the fviz_nbclust function from the factoextra (v1.0.7) R package to calculate the Within-Cluster Sum of Squares (WSS) for different numbers of clusters. WSS was used as a measure to assess the clustering performance. Next, the WSS values for all clusters were summed to obtain the Total Sum of Squares within Clusters (TSSE), and the TSSE was plotted against the number of clusters (k). Following the elbow method, we selected the optimal number of clusters by observing the TSSE curve. Subsequently, we used the Mfuzz R package[90] to perform the clustering and generate corresponding cluster plots, providing a visual representation of the clustering patterns of metabolites in different tissues. These clustering analyses allowed us to uncover the distribution patterns of metabolites across various samples and tissues, which

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provides valuable data support for subsequent functional studies and metabolic pathway inference.

Co-expression/Co-regulation Cluster Identification and Analysis

A total 313 metabolites were clustered using Mfuzz R package[90] into significant discrete clusters. After obtaining the average TPM value of each gene, Pearson correlation analysis was performed with the carpel metabolites. The genes with correlation of R > |0.85| with any of the metabolites were retained. A total 42,434 genes were retained. The 42,434 genes were clustered into six clusters using Mfuzz R package[90]. Visualization of all associations between genes and metabolites was carried out using Cytoscape software[91].

KEGG Enrichment Analysis and GO Analysis

For Metabolite KEGG data analysis, we used the web (https://www.metaboanalyst.ca/MetaboAnalyst/upload/ConvertView.xhtml) to change the metabolite ID to KEGG ID. FELLA R package was used for metabolite KEGG enrichment analysis[92]. Transcription GO term enrichment and KEGG term enrichment was performed using eggnog mapper[93] with the "-i wheat_V1.fasta -d virNOG -m diamond" options to get the codes of the corresponding genes in GO term and KEGG term. Applying a hypergeometric test with FDR correction (adjusted p value < 0.05), the analyses of GO term and KEGG term were conducted using the R package clusterProfiler[94].

Identification of candidate genes in starch and sucrose metabolism pathway

Starch and sucrose metabolism pathway (KEGG ID: ath00500) was identified in cluster9 through metabolic pathway enrichment analysis. Key regulatory enzymes

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involved in this pathway, including β-fructofuranosidase (EC 3.2.1.26), α-glucosidase (EC 3.2.1.20), and sucrose-phosphate phosphatase (EC 3.1.3.24), were retrieved from the KEGG database (https://www.genome.jp/kegg/). For each enzyme, the corresponding wheat gene indices (ats) were obtained from the "Gene" section of KEGG, and amino acid sequences were extracted from the linked webpages. These sequences were used for BLASTp analysis to screen candidate genes, with the highest alignment rate as the selection criterion. Among the identified candidates, *grain incomplete filling 1* (*TaGIF1*) was confirmed as a key gene involved in grain development.

Weighted gene co-expression network analysis

In the carpel samples, co-expression network modules were identified based on log2(TPM+1) value using WGCNA package in R[95]. Genes with a low coefficient of variation of count (CV < 0.25) among carpel samples were discarded and the remaining 9,636 genes were used for the analysis. Co-expression modules were obtained using the TOMtype "unsigned" and a Pearson correlation matrix was calculated between all pairs of selected genes from which adjacency matrix was constructed by raising the correlation matrix to the power of 12 (Fig. S10). Average linkage hierarchical clustering was carried out for grouping genes with highly similarity for co-expression patterns. The modules were identified using Dynamic Hybrid Tree cut algorithm[96], with the minimum module size to 50. The expression profile of each module was summarized by module eigen-gene defined as its first principal component, and modules with highly correlated (>0.8) eigen-genes were merged.

Exogenous hormone treatments and spike phenotype identification

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The hormones of 6-Benzylaminopurine (6-BA, 100mg/L), abscisic acid (ABA, 100uM), salicylic acid (SA, 100uM), 3-Indoleacetic acid (IAA, 100mg/L), methyl jasmonate (MeJA, 100uM) and gibberellic acid (GA, 100 μM) were used for treatments. 6-BA was first solubilized with 1% (w/v) NaOH, and then diluted with double-distilled H_2O to appropriate concentration[97]. ABA, IAA, MeJA and GA solubilized with 95% alcohol, and then diluted with double-distilled H_2O to appropriate concentration[98-101]. Tween 20 (0.01%, v/v) was added as a surfactant and mixed thoroughly. All the chemicals used were purchased from Sigma-Aldrich. The treatments were performed daily from inflorescences initiation to spikes maturation. The solutions were sprayed on the entire plants. The control lines were sprayed with the same volume of deionized water containing the same concentrations Tween-20 without hormones.

To evaluate the effects of exogenous hormone treatments on panicle development and yield-related traits, we conducted a systematic phenotypic assessment of spikes after the hormone application. The phenotypic measurements were performed at two levels:

Overall panicle trait measurement

Key agronomic traits of the entire spike were measured, including spike length, spike weight, total spikelet number, number of sterile spikelets at the top position, number of sterile spikelets at the bottom position, fertile spikelet number, and grain number. This analysis aimed to reflect the impact of hormone treatments on the general morphology and yield components of the spike, and therefore, the positional effects along the spike were not considered in this part.

Spikelet trait measurement at different positions

To further assess the direct effects of hormone treatments on spikelets from different positions within the spike, five fixed positions (e.g., Top, Up, Middle, Down,

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Bottom) were selected for representative spikelet sampling. For each sampled spikelet, the floret number per spikelet, grain number per spikelet, grain weight per spikelet, and spikelet weight were recorded.

In this analysis, comparisons were made between spikelets located at the same spike positions from hormone-treated plants and the corresponding water-treated controls. This design allows for the elimination of positional effects, thereby providing a more accurate estimation of the local phenotypic responses to hormone treatments.

Mutants

The *OPR3* gene, as previously described in the study[61], is utilized in our research to examine the effects of various wheat grain development models on grain growth. The information for the development of overexpression and RNAi lines were described in our previous work[61].

To get *GL-D1* overexpression plants, the coding sequences of *GL-D1* was individually amplified using DNA polymerase KOD FX (Toyobo, Osaka, Japan) from WT cDNA and cloned into the pMWB110 vector[102] using an In-Fusion Advantage PCR cloning kit (Clontech Laboratories, Takara, Tokyo, Japan). The resulting plasmids were transformed into the spring wheat cultivar Fielder via Agrobacterium-mediated transformation[103]. All primer sequences used in this study are listed in Table S39.

The Kronos (*Triticum turgidum,* tetraploid, AABB genome) *GL2* mutants mut-1 and mut-2 are both single mutants with mutations exclusively located on chromosome 6A. Wild-type seeds were treated with 0.7% EMS for 16 h at 25 °C to generate an EMS mutant library (http://wheatomics.sdau.edu.cn/). The *GL2* mutant was backcrossed with the wild-type Kronos and grown for two generations with no change in phenotype. *GL2* mutant was identified by PCR sequencing.

RT-qPCR Analysis

Fresh leaves of overexpression transgenic lines were collected and immediately frozen in liquid nitrogen. The samples were thoroughly grounded and transferred to

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centrifuge tubes. Total RNA was isolated with a Plant RNA Kit (TIANGEN, # DP441), first-strand cDNA was synthesized with a qPCR RT Kit (TOYOBO, #FSQ-101) according to the manufacturer's instructions. Real-time PCR analysis was performed using CFX real-time machine (Bio-Rad) and SYBR Premix Ex Taq Mix (Takara, #RR420A). Wheat *TaACTIN* was used as an internal control. The real-time PCR results were analyzed using CFX Maestro analysis software. Each experiment had three technical replicates. All primers used for RT–qPCR were listed in Table S39.

Haplotype analysis of *TaOPR3*, *GL1* and *GL2*

The SNPs in the 2 kb upstream and downstream region of *TaOPR3*, *GL1*, *GL2* were filtered by applying a MAF ≥ 5% cut-off and low-frequency haplotypes could arise owing to errors in SNP calling; we therefore filtered the haplotypes by retaining the ones with frequency ≥ 5. To analyze the variation of *TaOPR3*, *GL1*, *GL2*, we used 306 worldwide wheat accession, 172 hexaploidy accessions and 242 additional accessions (121 *Aegilops tauschii*, 121 tetraploid wheat varieties)[76,77], 829 Chinese wheat accessions released between 1900 and 2020[78,104], 355 Chinese and American wheat accessions[79]. Phylogenetic network were used popArt[105] to plot.

Accession Numbers

Sequence data from this article can be found in the EnsemblePlants database(https://plants.ensembl.org/Triticum_aestivum/Info/Index)under the following accession numbers: *TaGIF-D1* (*TraesCS2D02G293200*), *TaSPL16*(*TraesCS7A02G260500*, *TraesCS7B02G158500*, *TraesCS7D02G261500*), *TaOPR3* (*TraesCS7A02G412400*, *TraesCS7B02G311600*, *TraesCS7D02G405500*), *Grain Length1* (*TraesCS4D02G085000*), *Grain Length2(TraesCS6A02G416700)*_.

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AUTHOR CONTRIBUTIONS

Z.G. initiated the project. Z.G. and W.C. designed the experiments. Z.G., W.C., Y.H. and F.L. supervised the experiments. C.Y. and F.L. performed sequencing, and genomic-variant calls. Y.L., K.S. and Z.L. carried out transcriptome and metabolome analysis. Y.L. and L.Z. collected samples for transcriptome and metabolome analysis. A.Z. and W.C. conducted measurements of metabolites. L.S. carried out genotyping of NILs. J.Z., J.C., G.C., M.X. and Y.T. developed overexpression lines, RNAi lines and mutants. Z.W. and Z.S. collected phenotypic data of spike development. X.X., M S., Y.H. and Z.H. developed NILs. Y.L. and L.Z. plotted manuscript figures. All authors were involved in the writing and editing of the manuscript.

Figure legends

Figure 1. The high-spatiotemporal-resolution dissection of spike components in wheat. A, Spike dissection: spikelets and rachis from five positions across individual spike. B, Spikelet dissection: four florets at different positions within individual spikelet. C, Floret dissection: awn, palea, carpel, lemma, anther. D, Temporal dissection of carpel development: carpel form green anther stage (GrA) to 35 days post anthesis (DPA) stage. All of the above organ samples were taken at 12 stages from GrA to 35 DPA, except anthers from five stages from GrA to AN. A-D images were digitally extracted for comparison, and their scale bars were adjusted to be consistent for ease of comparison. The same images were used to show “carpel” in 1C and “AN” in 1D. E-H, Spike weight (E), spike length (F), grain weight (G)

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and chaff weight (H) at 12 stages form GrA to 35DPA stages. i, Spikelet weight of five positions from TP to 35DPA. J, The weight of the first floret from the bottom of the spikelet (F1 weight) from the spikelets at five positions during spike developmental process. K, The weight of four florets within individual spikelet at 10 stages from TP to 35DPA stage. GrA: green anther stage, YA: yellow anther stage, TP: tipping stage, HD: heading stage, AN: anthesis stage, DPA: days post anthesis.

Figure 2. Fig. 2. Summary of metabolome data of wheat spike. A, Overview of the 313 annotated metabolites. Different colors on the Y-axis represent different categories of metabolites. The X-axis represents different organs at 12 development stages. B, Principal component analysis (PCA) results of metabolites from 233 samples. C, Hierarchical clustering analysis results based on Pearson correlation coefficients of metabolites among 233 samples.

Figure 3. Global metabolite profiling of spike organs reveals common and organ-specific metabolic signatures. A, Fuzzy C-means clustering identifies nine different temporal patterns of metabolites. The x-axis represents 22 organs at 12 developmental stages. Different colors indicate different organs. Rachis and spikelet organs were taken at five positions (top, upper middle, middle, lower middle and bottom of spike); while the Y-axis represents log2-transformed, normalized Z-score values at each stage. "Membership" represents the degree of affiliation of metabolites to a specific cluster. B, The number of metabolites for the nine clusters (cluster1-9). ACD: amino acid derivatives, Fla: flavonoid, Lip: lipid, NAD: nucleic acid derivative, OrA: organic acid, Phe: phenolamide, Phy: phytohormone derivative, Plo: polyphenol, Sug: sugar, Va: vitamin. The number represent the number of the corresponding metabolites for each cluster. C, Metabolite enrichment patterns of cluster 5 for anther development at five stages (GrA, YA, TP, HD, AN). The colors represent the log2-transformed, normalized Z-score values in each stage, the redder the color the higher the enrichment. GrA: green anther stage, YA: yellow anther stage, TP: tipping stage, HD: heading stage, AN: anthesis stage. D, Anthers at five stages (GrA, YA, TP, HD, AN). The bar represents 1000 μm. E-F, Anther length (E) and width (F) at five stages (GrA, YA, TP, HD, AN). Data are presented as the mean (n = 10 plants). For box plots: the box represents the interquartile range (IQR), encompassing the first quartile (Q1), median, and third quartile (Q3); whiskers extend to values within 1.5 × IQR relative to the box boundaries; individual points denote outliers. Statistical analysis of significance was conducted via one-way analysis of variance (ANOVA), followed by the least significant difference (LSD) post hoc test. Different lowercase letters indicate statistically significant differences among groups ($P < 0.001$). G, Correlation between metabolites in cluster5 and anther

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length/width. The correlation coefficient is indicated by color: red hues (ranging from light to dark, corresponding to 0.3 – 0.9) represent positive correlations, while the color gradient from yellow to purple (0.0 – -0.3) represents negative correlations. The symbol "r" next to the color scale denotes the correlation coefficient, and the numerical labels clearly indicate the correlation coefficient values corresponding to each color interval.

Figure 4. Uneven distribution of hormones affects spike morphology. A, Fuzzy C-means clustering identified six temporal patterns of metabolites across five spikelet positions. Each line represents the expression pattern of metabolites assigned to a cluster. The X-axis represents individual samples, grouped by spikelet position and developmental stage. Samples 1-12 correspond to the top spikelet at 12 developmental stages, samples 13-24 to the upper middle spikelet at 12 stages, samples 25-36 to the middle spikelet at 12 stages, samples 37-48 to the lower middle spikelet at 12 stages, and samples 49-60 to the bottom spikelet at 12 stages. Different colors represent different spikelet positions (see legend). The Y-axis represents log2-transformed, normalized Z-score values for each sample. B, The number of metabolites assigned to each of the six clusters (Sp-cluster1-6). ACD: amino acid derivative, Fla: flavonoid, Lip: lipid, NAD: nucleic acid derivative, OrA: organic acid, Phe: phenolamide, Phy: phytohormone derivative, Plo: polyphenol, Sug: sugar, Va: vitamin. The number represent the number of the corresponding metabolites for each cluster. C, Network diagram of KEGG results of the six metabolite enrichment patterns. The color represents the KEGG -log10(*p* value). D, The significance of the differences among the five positions for each stage ($P < 0.05$, ANOVA). The red color indicates the significant differences of the hormone derivatives among the five positions at different stages. E, Representative images of wheat spikes from wild-type (WT) plants and plants subjected to exogenous hormone treatments (CK: cytokinin, ABA: abscisic acid, SA: salicylic acid, IAA: auxin, JA: jasmonate, GA: gibberellin). The images were digitally extracted for comparison, and their scale bars were adjusted to be consistent for ease of comparison. F-H, Spikelet dry weight (F), floret number per spikelet (G) and grain number per spikelet (H) across individual spike. The Y-axis represents the spike axis direction (from top to bottom), and the values on the axis correspond to the measured values of the respective traits for spikelets at different positions on a single spike. The X-axis represents the measured data of each trait, where the X-axis unit in Panel F is grams (g), and the X-axis unit in Panels G and H is "number". To more clearly display the characteristics of spike shape changes, we performed a mirroring process on the data to enhance the visual presentation effect of the patterns in spike shape changes.

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Figure 5. The regulation network of metabolome and transcriptome for carpel development in wheat. A, Dynamics of metabolites and gene expression during carpel developmental process. Fuzzy C-means clustering identified six different temporal patterns of carpel metabolite and gene expression. The x-axis represents carpel development stages, X-axis number indicates stages:1. GrA, 2. YA, 3. TP, 4. HD, 5. AN, 6. 5DPA, 7. 10DPA, 8. 15DPA, 9. 20DPA, 10. 25DPA, 11. 30DPA, 12. 35DPA. Y-axis represents log2-transformed, normalized Z-score values of metabolites and gene expression for each stage. B, The number of metabolites for the six clusters (cluster Ⅰ, Ⅱ, Ⅲ, Ⅳ, Ⅴ, Ⅵ). ACD: amino acid derivative, Fla: flavonoid, Lip: lipid, NAD: nucleic acid derivative, OrA: organic acid, Phe: phenolamide, Phy: phytohormone derivative, Plo: polyphenol, Sug: sugar, Va: vitamin. C, A network based on the correlations among genes and hormone derivatives in cluster Ⅲ. Pearson correlation coefficients were calculated between genes and metabolites in cluster Ⅲ. Pink and yellow colors represent genes and metabolites, respectively. The thickness of the line represents the size of R2. D, A network based on the correlations among metabolites and transcription factors (TFs). Pearson correlation coefficients were calculated for each TF and metabolite in six clusters. Red and blue lines indicate positive and negative correlations, respectively. Circle and triangle represent metabolites and genes, respectively. Different colors of circles indicate different classes of metabolites.

Figure 6. *TaOPR3*, *GL1*, and *GL2* are associated grain size. A, Genes related to jasmonic acid synthesis pathway in cluster VI. B, C, Grains of KN199 (wild type), *TaOPR3* RNAi lines (RNAi-1, RNAi-2, RNAi-3) and overexpression (OE) lines (OE-1, OE-2, OE-3). D, E, Grain length and width of KN199 (wild type), *TaOPR3* RNAi and OE lines. F, G, Grains of Fielder (wild type) and *Grain Length 1* (*GL1*) overexpression lines (F.OE-1, F.OE-2, F.OE-3,F.OE-4). H, I, Grain length and grain width of Fielder (wild type) and overexpression lines (F.OE-1, F.OE-2, F.OE-3, F.OE-4). J, K, Grains of Kronos (wild type) and *Grain Length 2* (*GL2*) mutants (mut-1, mut-2) lines. L, M, Grain length and grain width of Kronos (wild type) and *Grain Length 2* (*GL2*) mutants (mut-1, mut-2) lines. All statistical analyses in this figure were performed using the following methods: data are mean (n = 10 plants). Data were analyzed using the t-test function. Significant differences were determined at * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$ levels compared with wild type and mutant or overexpression lines.

Figure 7. Geographical differentiation, breeding selection, and domestication for the haplotypes of *TaOPR-A3*. A, The four major haplotypes of *TaOPR-A3*. Alleles that

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are identical to or different from the ones in the Chinese Spring (IWGSC RefSeq v1.0 reference) genome are indicated by purple and red bars, respectively. B, Geographical distribution of the four haplotypes (n = 86 accessions [Hap-A1$^{TaOPR-A3}$], 165 accessions [Hap-A2$^{TaOPR-A3}$], 18 accessions [Hap-A3$^{TaOPR-A3}$]), 31 accessions [Hap-A4$^{TaOPR-A3}$]) based on worldwide wheat accessions. AF, Africa; AS, Asia (except Middle East); EU, Europe; ME, Middle East; NoA, North America; OA, Oceania; SA, South America. C, Thousand grain weight for accessions carrying each of the four haplotypes (n = 86 accessions [Hap-A1$^{TaOPR-A3}$], 165 accessions [Hap-A2$^{TaOPR-A3}$], 18 accessions [Hap-A3$^{TaOPR-A3}$]), 31 accessions [Hap-A4$^{TaOPR-A3}$]) of *TaOPR-A3*. The significance analysis was performed according to one-way analysis of variance (ANOVA) and least significant difference test, LSD. Different letters indicate significant differences ($P < 0.01$). Values are means+SD; D, Frequency of four haplotypes among 829 Chinese wheat accessions released from 1900 to 2020. E, Frequency of four haplotypes among 352 Chinese and USA wheat accessions. F, Haplotype networks for *TaOPR-A3*. The haplotype networks were developed based on the SNPs in high linkage disequilibrium in a haplotype using 124 hexaploidy accessions and 148 additional accessions (91 Aegilops tauschii, 57 tetraploid wheat varieties).

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COMPETING INTERESTS

The authors declare no competing interests.

AVAILABILITY OF DATA AND MATERIALS

The genotype of 306 wheat accession used in this study have been deposited in the Genome Variation Map (https://bigd.big.ac.cn/gvm) under accession number GVM000463. The sequence data of the 355 wheat accessions used in this study were

project CRA005878. The genotype data in variant call formats (VCF) are available in the NGDC database under accession number GVM000315. The carpel RNA-seq raw sequence data reported in this paper have been deposited in the Genome Sequence Archive[106] in National Genomics Data Center[107] China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: CRA013211) that are publicly accessible at https://ngdc.cncb.ac.cn/gsa. Two replicates of the metabolomic data are available in Supplementary Table S40 and the raw datas have been deposited in the Open Archive for Miscellaneous Data in National Genomics Data Center China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (OMIX011559).

CODE AVAILABILITY

The custom code for RNA-seq and metabolism analysis are available at https://github.com/halo9369/Yangyang-Liu.

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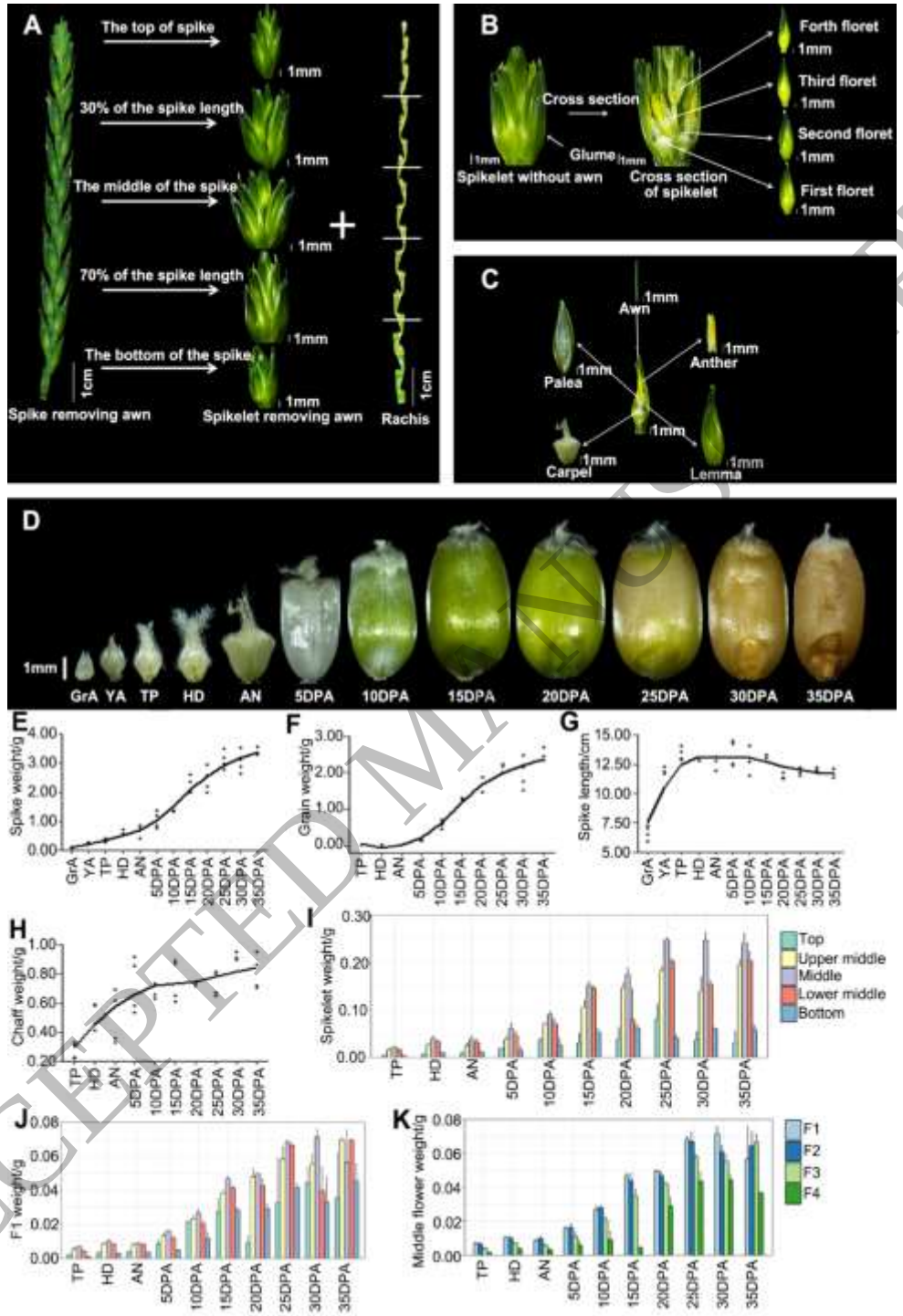

Figure 1

170x244 mm (x DPI)

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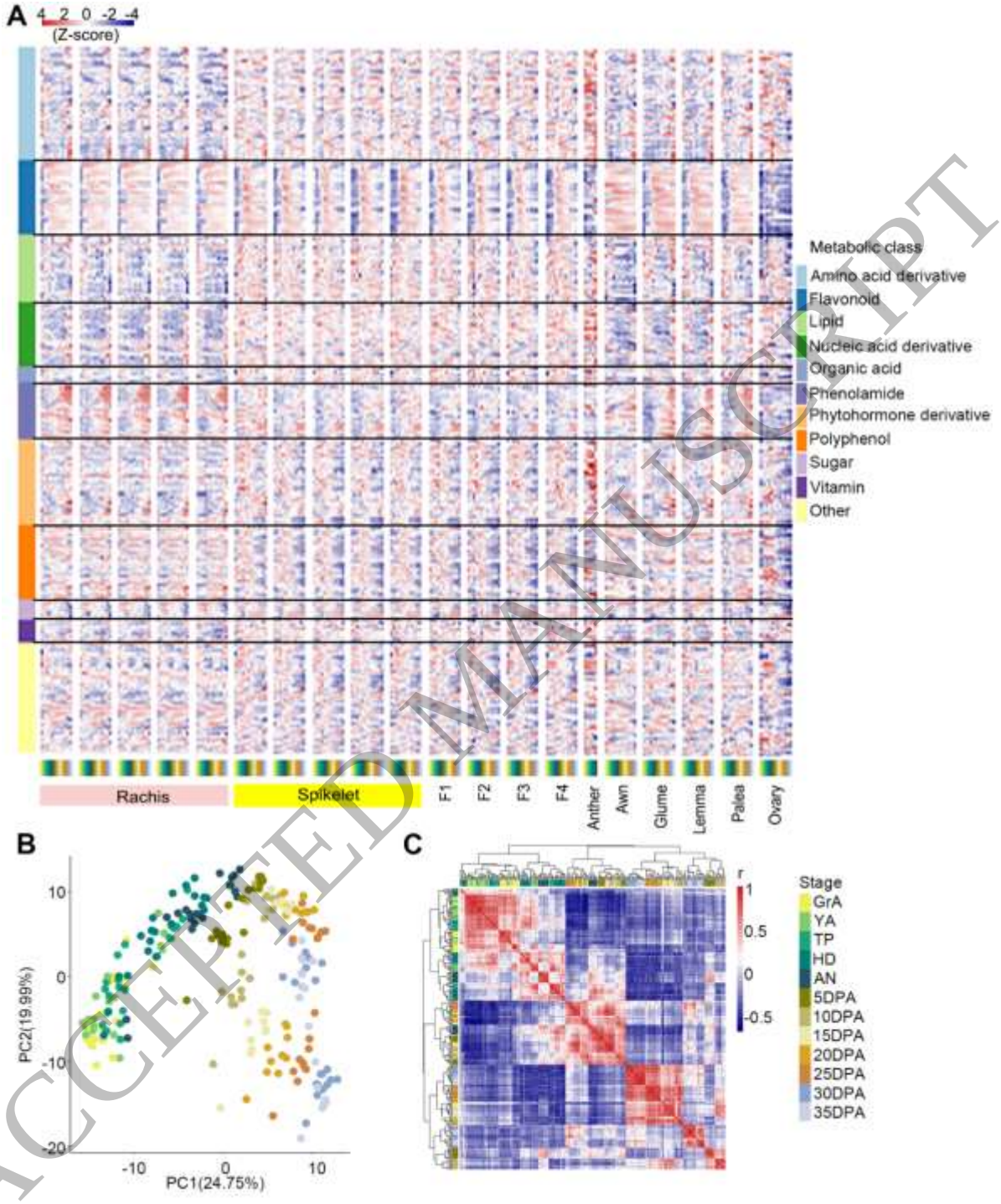

Figure 2

171x209 mm (x DPI)

ACCEPTED MANUSCRIPT

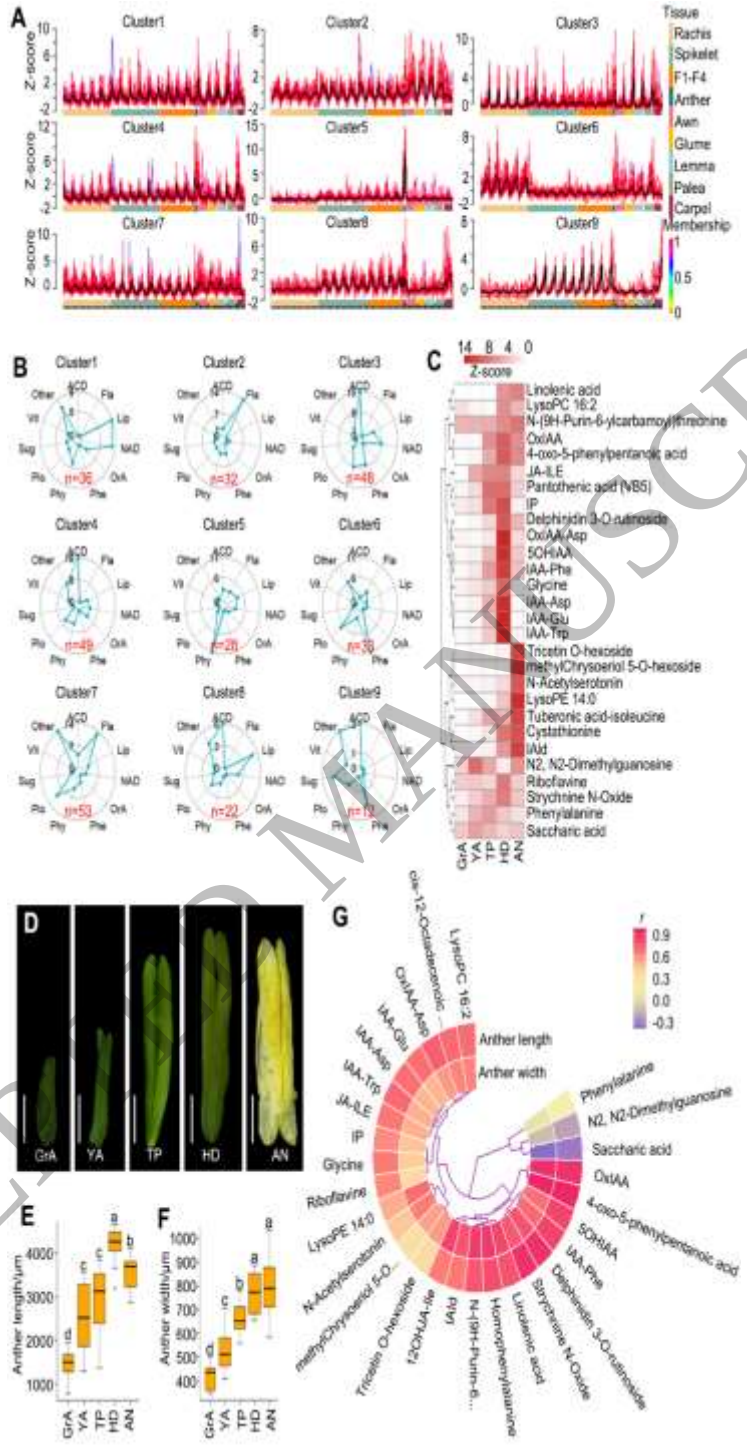

Figure 3

170x257 mm (x DPI)

ACCEPTED MANUSCRIPT

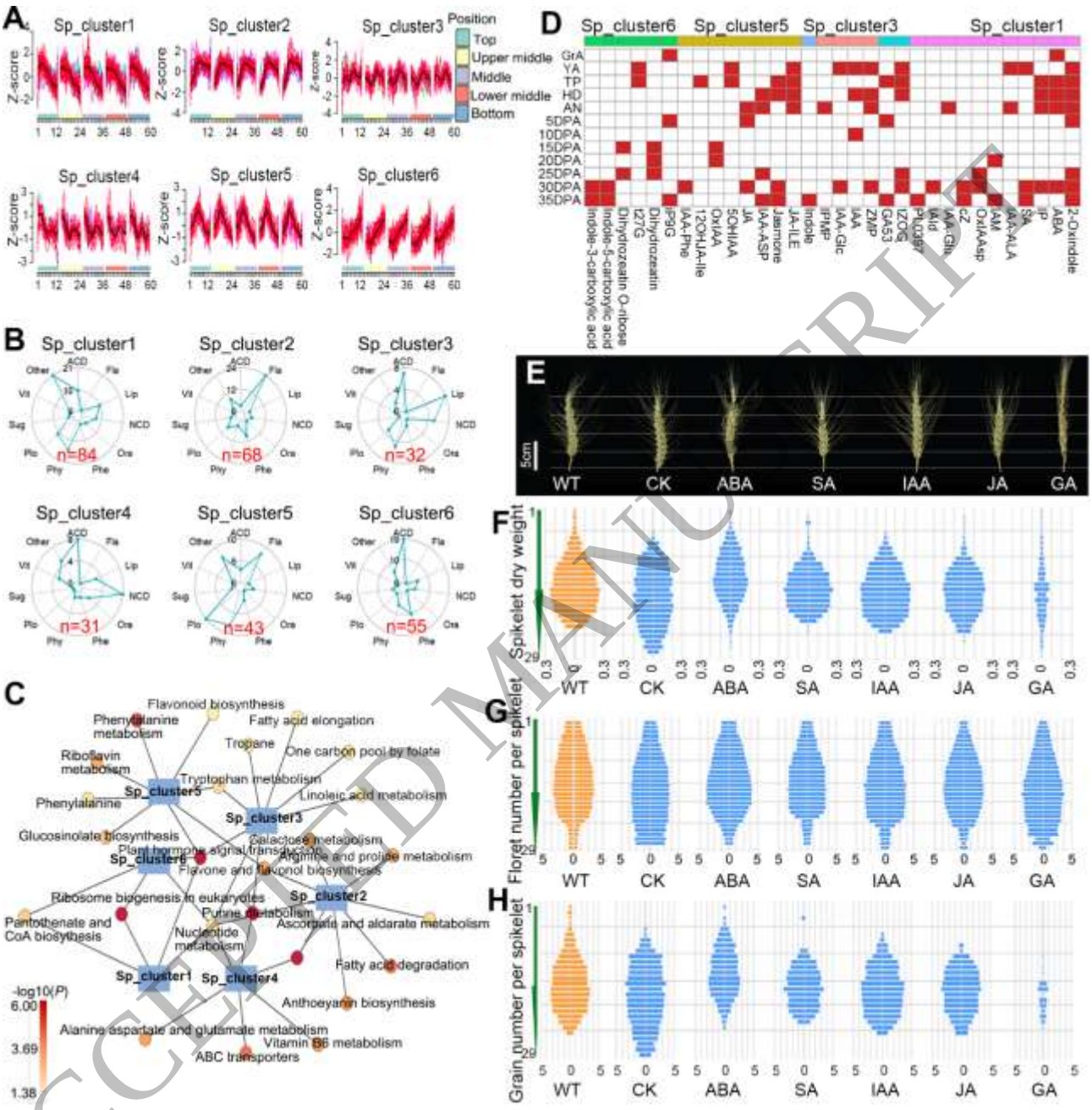

Figure 4

170x172 mm (x DPI)

ACCEPTED MANUSCRIPT

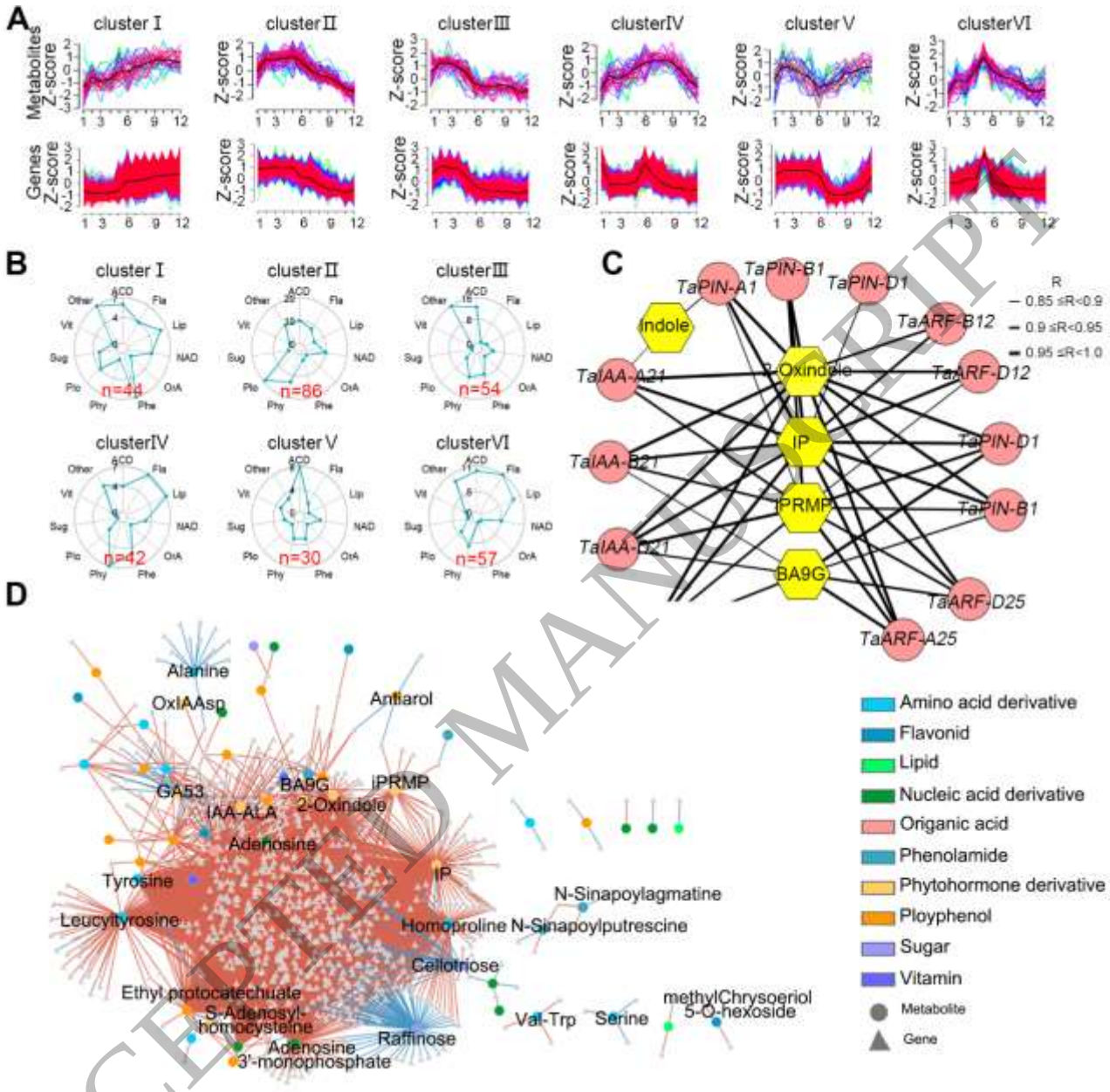

Figure 5

170x167 mm (x DPI)

ACCEPTED MANUSCRIPT

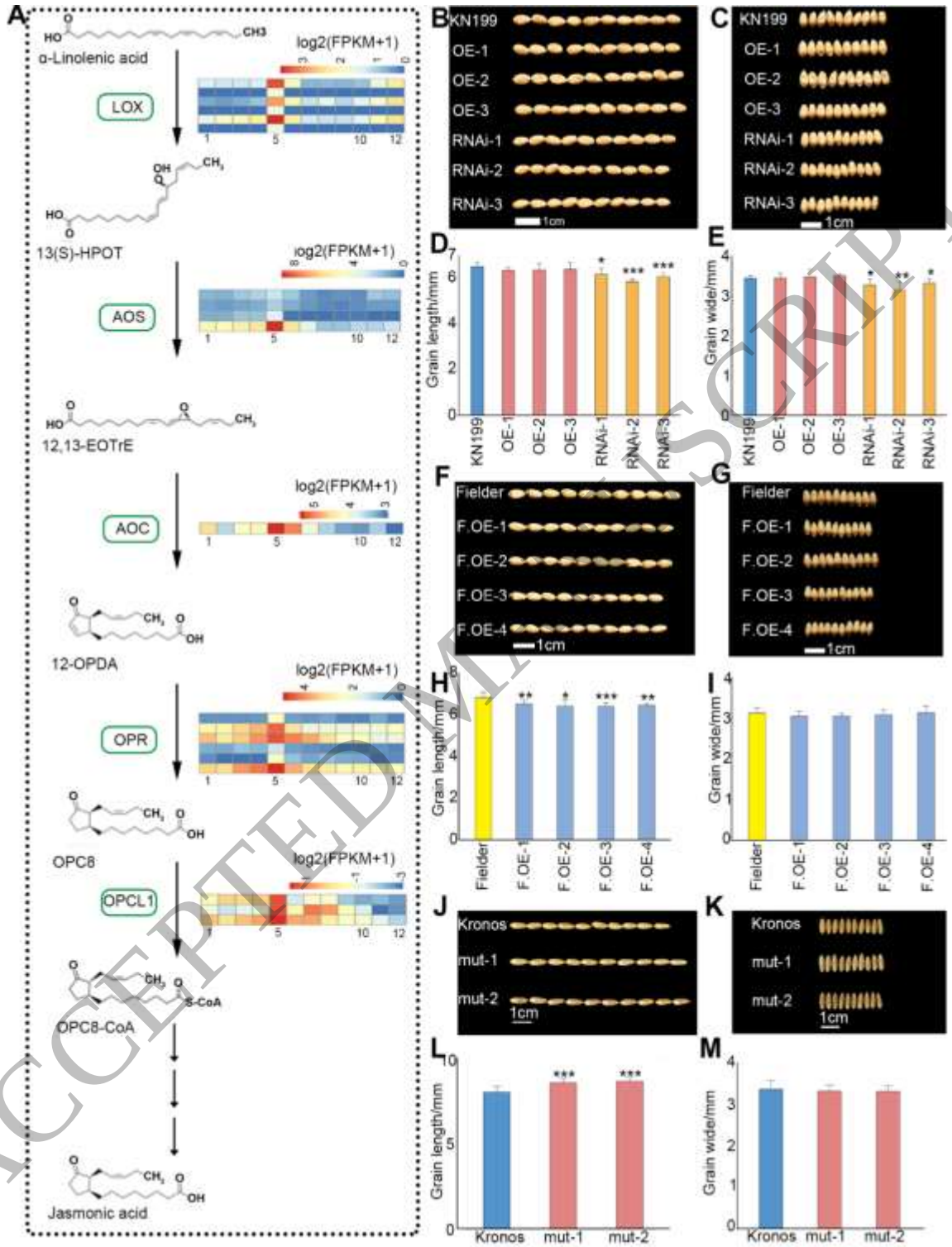

Figure 6

170x220 mm (x DPI)

ACCEPTED MANUSCRIPT

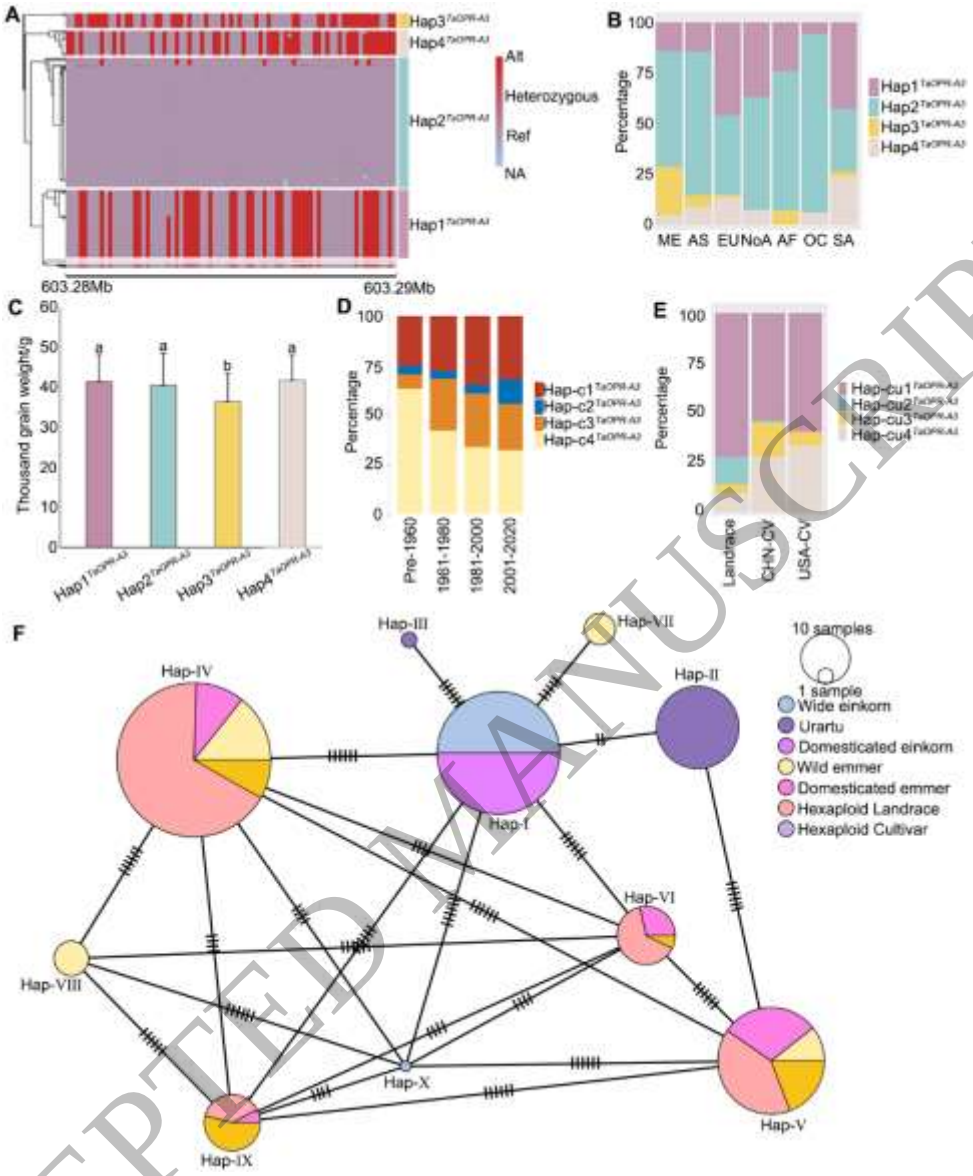

Figure 7

128x155 mm (x DPI)

ACCEPTED MANUSCRIPT